

CODE SPORT



In this month's column, we feature a bunch of interview questions based on data science.

Since quite a few of our student readers have requested me to feature data science interview questions, let's look at a few of them in this month's column.

1. Given that most data science folk use Python extensively, if you are preparing for data science interviews, it would be good to brush up on your Python coding skills. It is also important to understand and be able to explain the advantages and disadvantages of using an interpreted language such as Python as opposed to using a natively compiled language like C. Frequently, a candidate is asked to write Python code to generate the prime numbers that lie between 1 and 10,000. You are first asked to write a single threaded version of the code. Then you are asked if you can come up with a multi-threaded version of the code which is faster, given that you are running your code on a multi-core machine. Here is where the tricky part of the question lies. Given that you have written multi-threaded code in Python to compute all the prime numbers that lie between 1 and 10,000, to run it on a 4-core machine, how much faster do you expect it to be, compared to the single-threaded machine? The clue is to think about CPython's Global Interpreter Lock. As a bonus question, the interviewer could ask you, "Given that your favourite language is still Python and hence you would not like to rewrite the multi-threaded version in C, how can you make sure that your multi-threaded Python code is actually faster than the single-threaded code on the multi-core machine?"
2. You are given a dataset containing the weekly rainfall experienced in Bengaluru for the last 20 years. The data contains the week, date, and the corresponding rainfall in millimetres for that week. Your first task is to find out the outliers in your data. How would you do this?
3. You are given a set of documents and you are asked to perform clustering based on document level similarities. You are familiar with different clustering algorithms such as 'kmeans', so you are confident that you can do this. However, the interviewer now adds a second part to the problem — he wants you to label the clusters. How would you generate labelled clusters? What are the means of generating meaningful labels for the clusters? Can the clusters be labelled post the clustering process, prior to it or during the process itself? Describe your solution and explain why you made that particular choice?
4. Consider the problem mentioned in (3). Given a set of documents, you first need to calculate the similarity between the documents. Let us assume that you decided to use a bag-of-words model. What are the similarity measures that you can use? Under what circumstances would you use the cosine similarity measure and when would you select the Jaccard similarity measure?
5. You are given a document corpus consisting of more than a 100,000 documents, with each document containing at least 1000 words. You decide to construct a Term Document Frequency matrix M where each row represents a term in your vocabulary and each column represents a document. Given a document D , you are asked to find all documents that are similar to D . How would you go about it? Given that M is a high-dimensional sparse matrix, how would you perform dimensionality reduction on M ?
6. Consider the problem mentioned in (5). Given the document corpus, you will be given a query where you need to retrieve the top K documents which are most relevant to that query. How would you leverage your solution for (5) to solve this problem?